

EN401 — Handout 2: Energy Chain Data Structures

Activity Ratios, Efficiencies, and Commodity Balances

1. Data Structure Overview

Handout 2 parameterizes the energy network using InputActivityRatio and OutputActivityRatio to establish strict physical conservation of energy across every transformation stage.

2. Plant Efficiencies & Activity Ratios

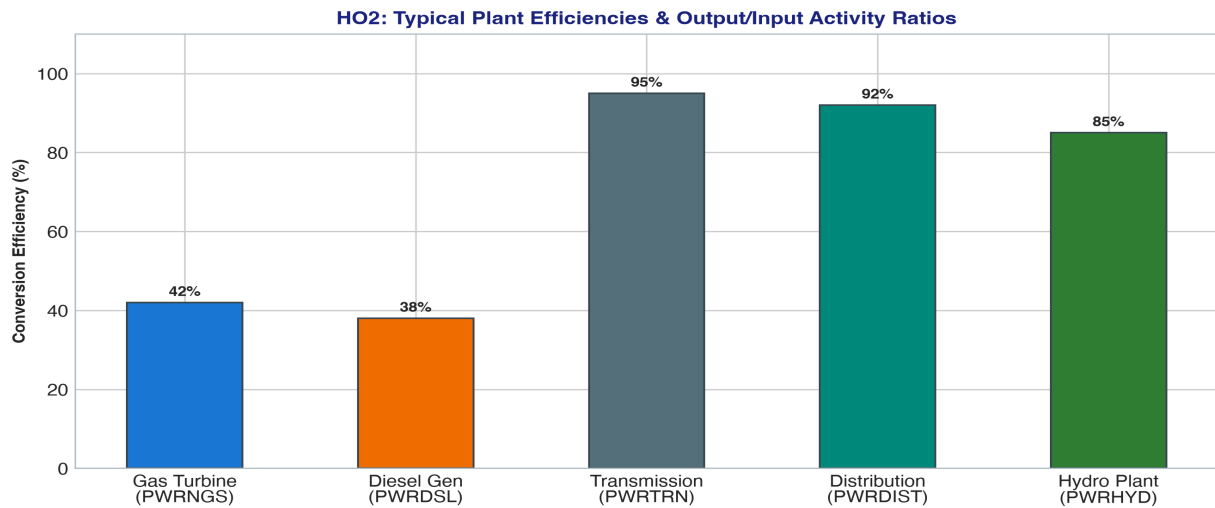


Figure: Thermal and network efficiencies across conversion technologies.

3. Core Takeaways

- Defined InputActivityRatio to model thermal losses in combustion technologies.
- Established delivery network efficiencies: 95% for transmission and 92% for distribution.